William L. Spies II

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SKILLS, STANDARDS, AND TOOLS

C++11/14/17	Git	ePlan Electric P8	ANSI/RIA R15.06 Experience
Python 3	CMake	AutoCad Electrical	NFPA 70E Experience
ROS (Kinetic, Melodic)	Google Test	Altium Designer	ISO 13849 Experience
Boost, Eigen	Catch2	LTSPICE	ISO 26262 Experience
OpenCV, PCL	Jira, Confluence, Bitbucket	MathCAD	Budget & Labor Estimation
Numpy	Word, Excel, Project	Fanuc HandlingTOOL	Leadership Development
CUDA	PowerPoint, Visio	Fanuc iRVision	

HISTORY

Outrider Technologies, Inc. — Golden, CO

Robotics Software Engineer — March 2019 - Present

- Key software engineer on the Outrider TrailerConnect team.
- Designed and authored ROS and native C++ code for TrailerConnect™ Obstacle Detection / Obstacle Avoidance (ODOA) system, incorporating off-the-shelf SICK and FLIR MCS components.
- Performed Machine Safety Risk Assessment (compliant with R15.06) on the full TrailerConnect robotic system and validated use of SIL3/Cat4 components and robotic arm controller features to mitigate risks to property and personnel.
- Contributes upgrades and patches to high-level decision and execution engine in C++.
- Responsible for commissioning of Yaskawa GP7 robot arms, integrated Intel RealSense cameras, SICK safety components, embedded computers, and other devices used throughout the TrailerConnect™ system.
- With input from Chief Sales Officer, pioneered data collection and analysis system for understanding TrailerConnect system performance in relation to business and engineering objectives.
- Participates in Agile sprint planning, task time estimation, and task dependency organization.
- Participates in code reviews of TrailerConnect software, as well as software written by other Outrider engineering teams.
- Periodically assists System Integration & Test team with designing automated TrailerConnect™ tests.

Northwestern University — Evanston, IL

Research Scientist, Computational Photography Lab — June 2018 - December 2018

Worked under Dr. Oliver Cossairt to assist with numerous active lab projects:

- Spearheaded joint effort with Dr. Oliver Cossairt and NU ACCESS to leverage Python, WebRTC, and consumer tablet devices for 3D surface scanning using structured light projection and deflectometry. Patent pending.
- Spearheaded joint effort with Dr. Florian Willomitzer to adapt Flying Triangulation (**arxiv:1305.4168**) from MATLAB source code to modern C++, Eigen, OpenCV, and PCL for real-time operation.

Master of Science, Robotics — September 2017 - September 2018

Graduate coursework exposure and independent project focus:

- Robotic uses of computer vision (e.g. visual odometry, 3D reconstruction, camera calibration).
- Embedded systems programming in modern C++ (11/14/17) and Python, including heavy usage of ROS.
- Implementing real-time kinematic control of autonomous robotic systems.
- Applications of machine learning for object detection / classification, and image processing.
- General purpose GPU programming with CUDA.
- Custom PCB design and fabrication, and advanced mechanical fabrication (laser cutting and 3D printing).

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ATS Automation — Lewis Center, OH

Systems Project Engineer; Laser Safety Officer — February 2017 - August 2017

- Orchestrated team of 12 engineers with more than 100 collective years of PLC, HMI, and robot programming experience on industrial automation projects for First Solar, General Motors, Tesla, Oral-B, Doosan, and other customers.
- Aggregated project status and reported engineering progress to division leadership on a weekly basis.
- Assembled project task lists, designed high-level architecture for early-stage projects, anticipated manpower needs, and coordinated with other engineering disciplines on an as-needed basis.
- Developed material and labor estimates (scope up to 10,000 man-hours) for sales department on an as-needed basis.
- Outfitted new vision lab for the development of novel machine vision strategies for future projects and sales.
- Coordinated with sister ATS divisions on global code standard development for use across entire organization.
- Continued Laser Safety Officer supervisory responsibilities.

Systems Design Engineer; Laser Safety Officer — September 2015 - February 2017

- Designed, implemented, and validated machine controls as lead programmer on industrial automation projects for First Solar, Doosan, 3M, and other customers over an 18-month period.
- Developed software architecture, machine control code, safety systems, data handling infrastructure, and human-machine interfaces per individual customer requirements.
- Programmed and validated Fanuc industrial robots for material handling and machine vision applications.
- Coordinated project schedules, task lists, and action items between project team members.
- As Laser Safety Officer, handled matters of safe laser integration, operation, and maintenance through authoring workspace policies and performing optical calculations relevant to Class 3/4 laser safety.

Electrical Design Engineer — June 2012 - September 2015

- Designed industrial automation systems for custom automated manufacturing systems, metrology systems, and material handling systems for Dow Chemical, General Motors, TRW, and other customers over a three-year period.
- Served as lead controls engineer for General Motors EV/HEV battery assembly systems with electrical budgets exceeding $2 million USD.
- Participated in design reviews with all customers as part of formal design approval processes.
- Coordinated with project-specific assembly teams for install supervision and validation of control systems prior to and after system power-up.
- Participated in validation of safety systems in compliance with ISO 13843 and ANSI/RIA R15.06 safety standards.

University of Cincinnati — Cincinnati, OH

Bachelor of Science, Electrical Engineering — June 2007 - June 2012

Minor in Photonics

Undergraduate coursework exposure and independent project focus:

- Primary focus included FPGA, microcontroller, solid-state electronics design, and optical communication.
- Secondary focus included microwave communication, semiconductor physics, thermodynamics, and orbital mechanics.
- Leadership involvement in University IEEE chapter and College of Engineering Tribunal body.
- Involved in community service and leadership development at the Center for Community Engagement.

References Available Upon Request